Directions: Answer the following questions and circle the corresponding option.

$$8, -4, 2, -1, \ldots$$

1. The first four terms of the geometric sequence $g_n = g_0(r)^n$ are shown above. What is the value of r?

(a) $r = 2$ (b) $r = -2$ (c) $r = \frac{1}{2}$ (d) $r = -\frac{1}{2}$

$$r = \underbrace{\frac{-4}{8}}_{-\frac{1}{2}} = \underbrace{\frac{2}{-4}}_{-\frac{1}{2}} = \underbrace{\frac{-1}{2}}_{-\frac{1}{2}} = -\frac{1}{2}$$

2. Let a_n be an arithmetic sequence where $a_3 = 5$ and $a_9 = 14$. What is the value of a_{12}?

(a) 18 (b) 18.5 (c) 19 (d) 23

$a_{n+1} = a_n + k$
$a_4 = a_3 + k = 5 + (4-3)k$
$a_5 = a_4 + k = 5 + k + k = 5 + 2k = a_3 + (5-3)k$
$a_6 = a_5 + k = 5 + 2k + k = 5 + 3k = a_3 + (6-3)k$
$\vdots$
$a_9 = a_3 + (9-3)k = 14$

$5 + 6k = 14$
$6k = 9$
$k = \frac{9}{6} = 1.5$
$a_{12} = a_9 + 1.5(12-9) = 14 + 1.5(3) = 18.5$

3. Let g_n be a geometric sequence where $g_4 = -5$ and $g_6 = -45$. Which of the following could be r?

I. $r = -3$ II. $r = 3$ III. $r = 9$

(a) I only (b) II only (c) III only (d) I and II

$g_6 = g_4 r^2$
$-45 = -5r^2$
$r^2 = 9 \Rightarrow r = \pm 3$

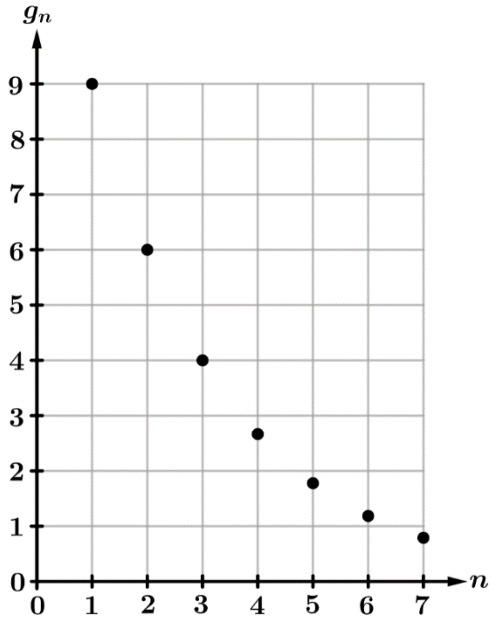
4. Let a_n and g_n be an arithmetic and geometric sequence with $a_1 = g_1 = -1$ and $a_4 = g_4 = 8$. What is the value of $a_5 - g_5$?

(a) -5 (b) 0 (c) 5 (d) 27

$a_{n+1} = a_n + k$
$a_4 = a_1 + 3k$
$a_4 = -1 + 3k = 8$
$-1 + 3k = 8$
$3k = 9 \Rightarrow k = 3$

$g_{n+1} = g_n r$
$g_4 = g_1 r^3$
$8 = -1r^3$
$-8 = r^3$
$-2 = r$

$a_5 = a_4 + 3 = 8 + 3 = 11$
$g_5 = g_4(-2) = 8(-2) = -16$
$a_5 - g_5 = 11 - (-16) = 27$



5. Values of the terms of a geometric sequence g_n are graphed in the figure above. Which of the following is an expression for the nth term of the geometric sequence?

(a) $g_n = 4\left(\frac{3}{2}\right)^{n-2}$ $g_1 = 4\left(\frac{3}{2}\right)^{1-2} = 4\left(\frac{3}{2}\right)^{-1} = 4\left(\frac{2}{3}\right) = \frac{8}{3} \neq 9$

(b) $g_n = 6\left(\frac{2}{3}\right)^{n-2}$ $g_1 = 6\left(\frac{2}{3}\right)^{1-2} = 6\left(\frac{2}{3}\right)^{-1} = 6\left(\frac{3}{2}\right) = \frac{18}{2} = 9$

(c) $g_n = 9\left(\frac{2}{3}\right)^{n}$ $g_1 = 9\left(\frac{3}{2}\right)^{1} = 9\left(\frac{3}{2}\right)^{-1} = \frac{27}{2} \neq 9$

(d) $g_n = 9\left(\frac{3}{2}\right)^{n-1}$ $g_1 = 9\left(\frac{3}{2}\right)^{0} = 9$ $g_2 = 9\left(\frac{3}{2}\right)^{2-1} = 9\left(\frac{3}{2}\right) = \frac{27}{2} \neq 6$